

The Hypothalamus



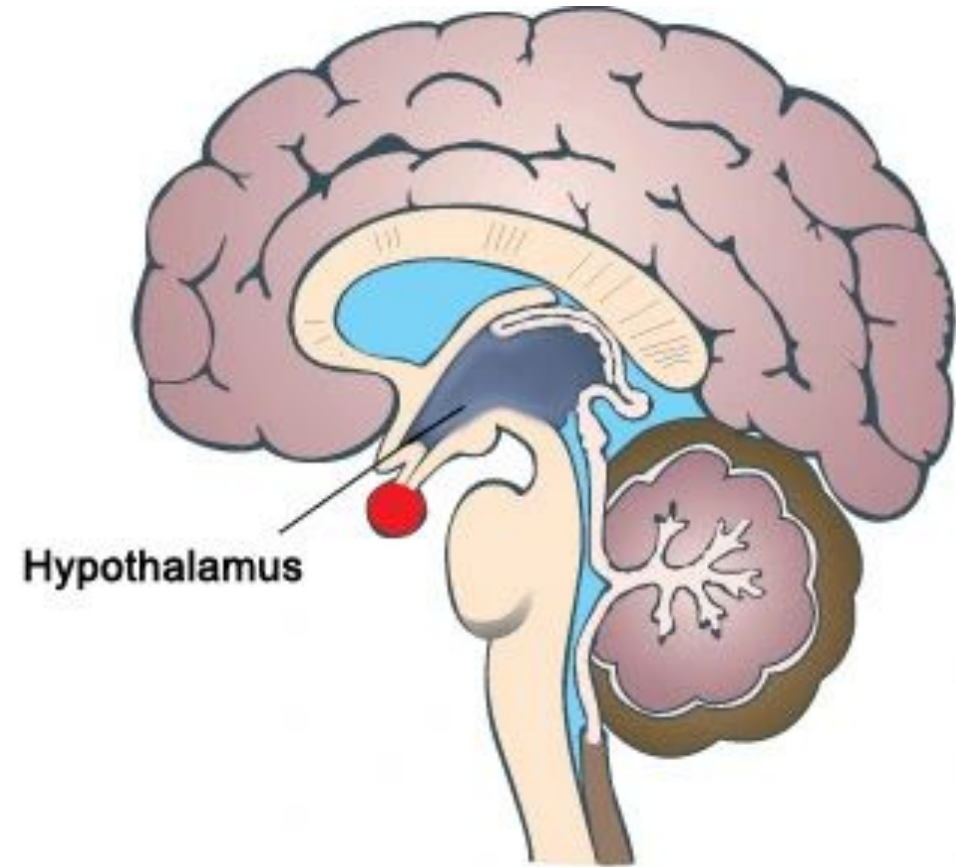
Learning objectives:

At the end of the lecture, the students should be able to:

- **Define hypothalamus & its functional significance.**
- **Describe its extent and relations.**
- **Enlist the nuclei of medial & lateral zones and the functions of hypothalamus.**

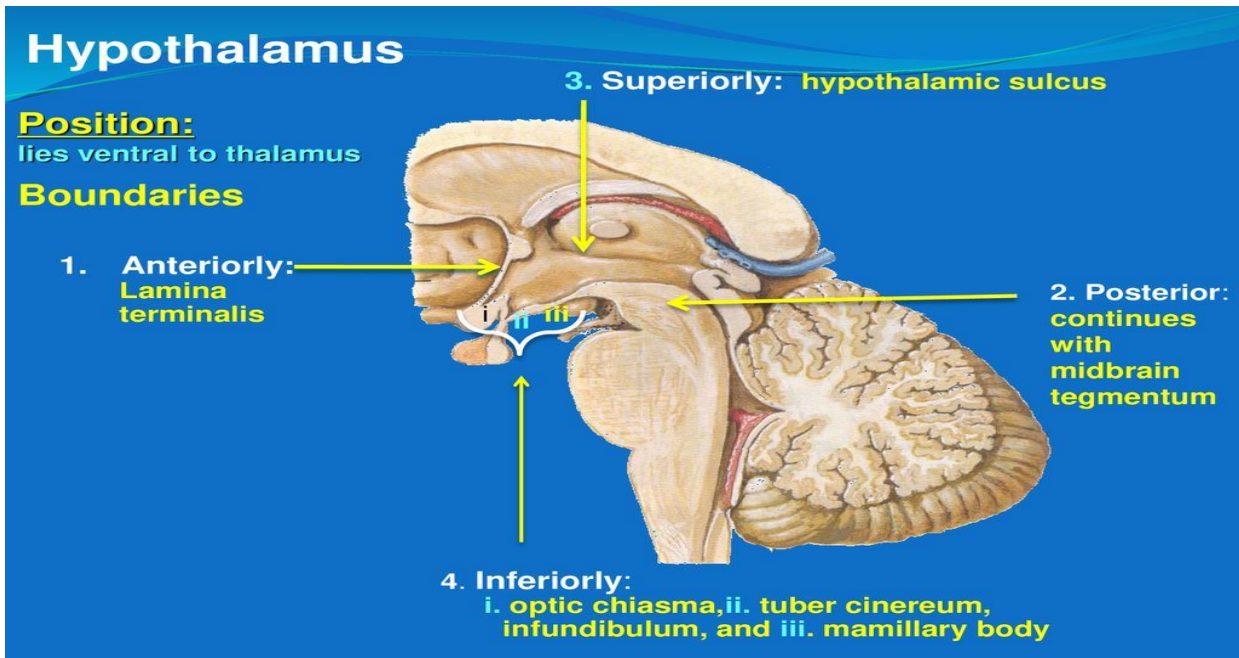
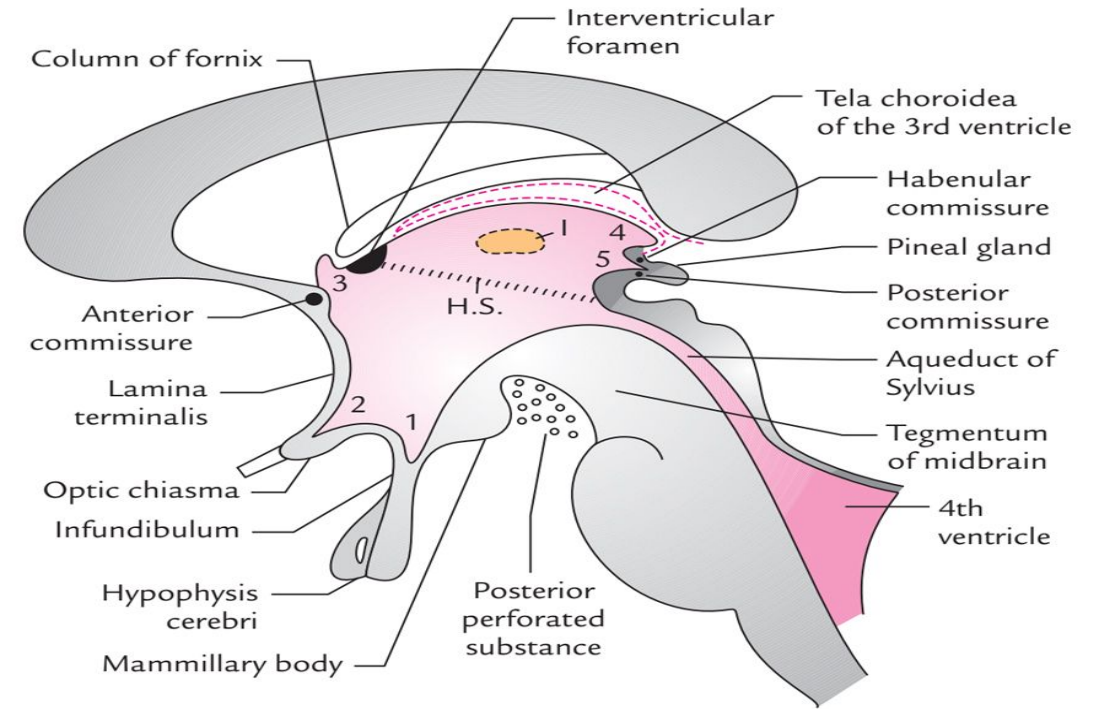
Introduction:

- Constitutes 0.3% of total brain.
- It is an important part of central nervous system which is essential for life.
- It controls autonomic nervous system and endocrine system, thus indirectly controls body homeostasis.
- Physiologically, there is hardly any activity in the body that is not influenced by hypothalamus.
- Microscopically, it is composed of small nerve cells arranged in groups/nuclei.



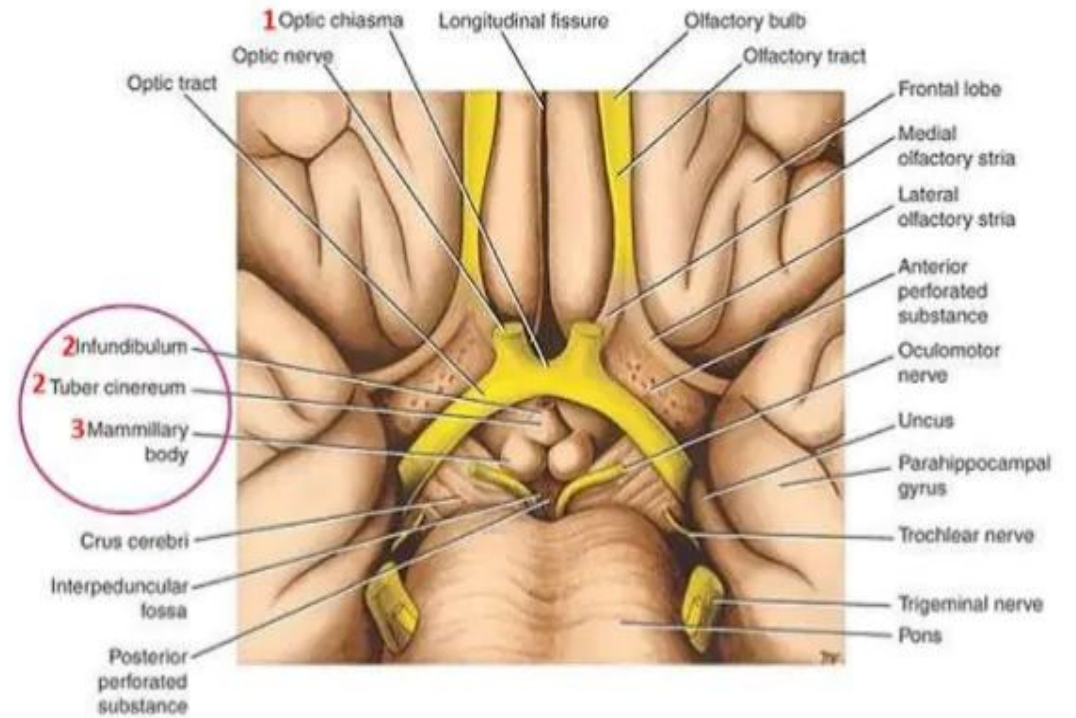
Extent and relations:

- Hypothalamus lies below hypothalamic sulcus.
- It extends from optic chiasma to caudal border of mamillary bodies.
- It forms floor and inferior part of lateral walls of third ventricle.
- **Preoptic area:** lies anterior to hypothalamus and functionally included in it. It extends from optic chiasma to linea terminalis & anterior commissure.
- Caudally, it merges into tegmentum of midbrain.
- Lateral boundary is formed by internal capsule.



- From below, the hypothalamus is seen to be related with the following structures from anterior to posterior:
 - Optic chiasma:** is a midline structure where medial fibers of optic nerve cross to continue as optic tracts.
 - Tubercinerium:** is a hollow eminence of gray matter located between optic chiasma & mamillary bodies. It is a part of hypothalamus, &
 - Infundibulum:** is a funnel-shaped extension of hypothalamus which connects pituitary gland to the base of brain.
 - Mamillary bodies:** two small hemispherical bodies located side by side posterior to tubercinereum. They have a central core of gray matter invested by a capsule of myelinated nerve fibers.

Inferior Relation



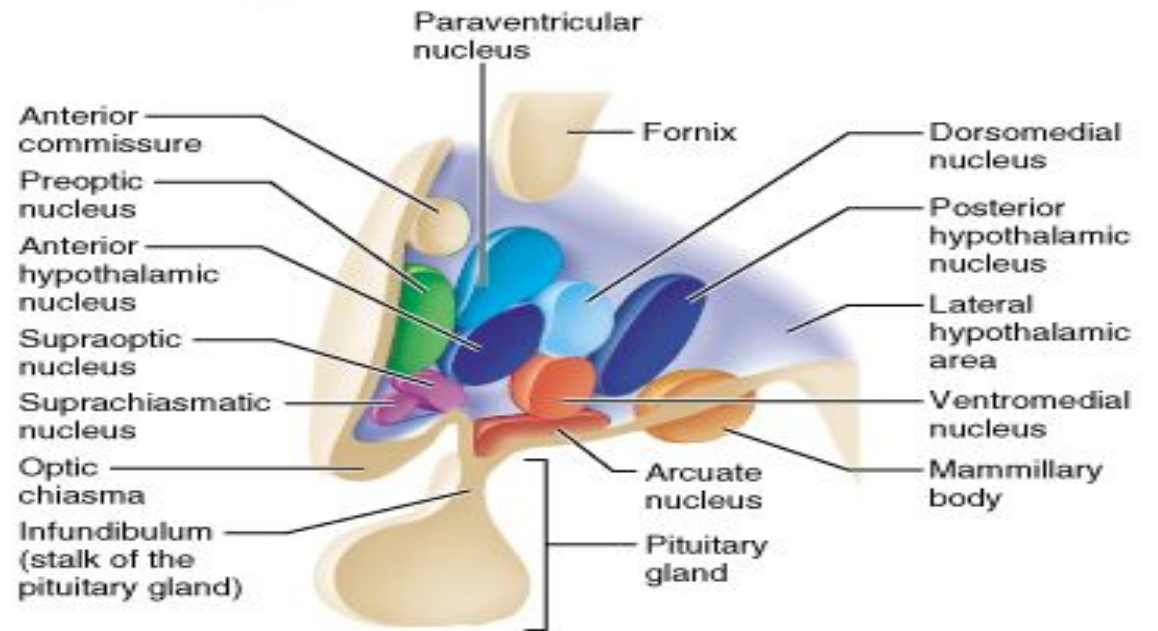
Structures related to hypothalamus at the base of the brain

Hypothalamic Nuclei:

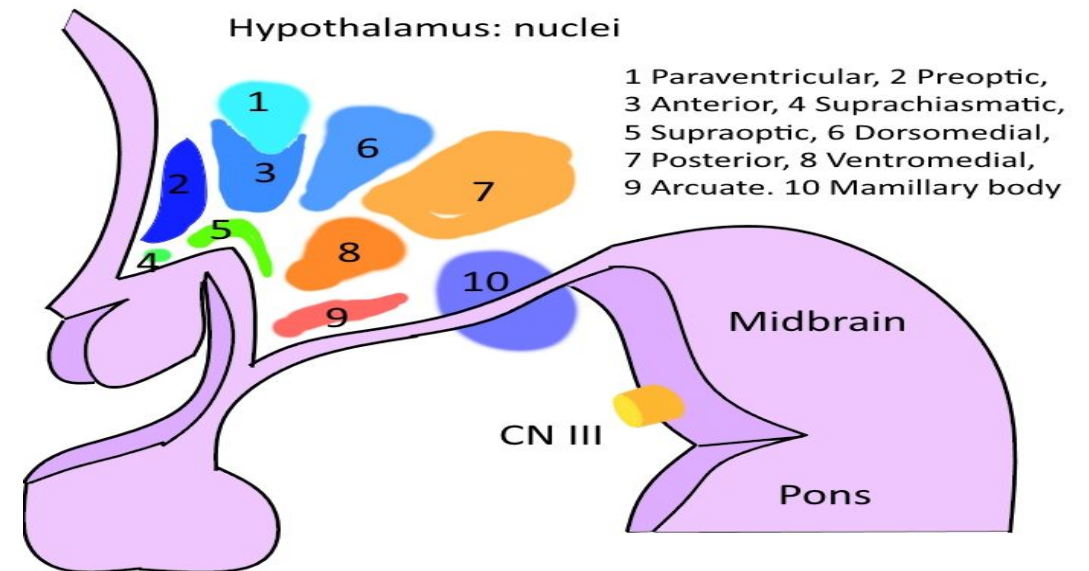
- For descriptive purposes, nuclei of hypothalamus are divided by an imaginary paramedian plane into medial & lateral zones.

- **Nuclei of medial zone (anterior to posterior):**

1. Part of preoptic nucleus
2. Anterior nucleus
3. Part of suprachiasmatic nucleus
4. Paraventricular nucleus
5. Dorsomedial nucleus
6. Ventromedial nucleus
7. Infundibular/arcuate nucleus
8. Posterior nucleus



(b)



Nuclei of lateral zone (anterior to posterior):

1. Part of preoptic nucleus
 2. Part of suprachiasmatic nucleus
 3. Supraoptic nucleus
 4. Lateral nucleus
 5. Tuberomammillary nucleus &
 6. Lateral tuberal nuclei
- The hypothalamus receives information from the whole body through nervous connections, blood stream, & cerebrospinal fluid.

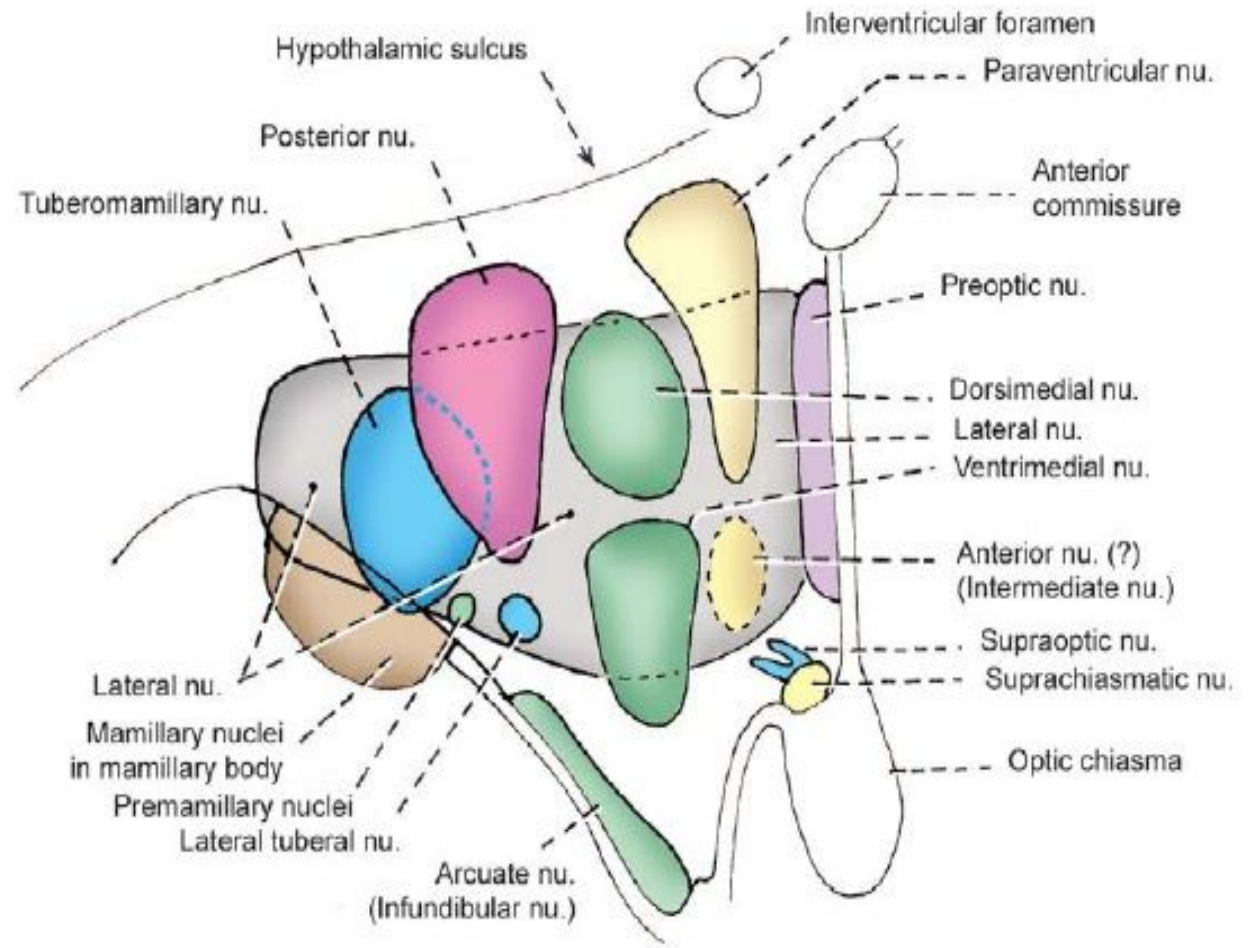
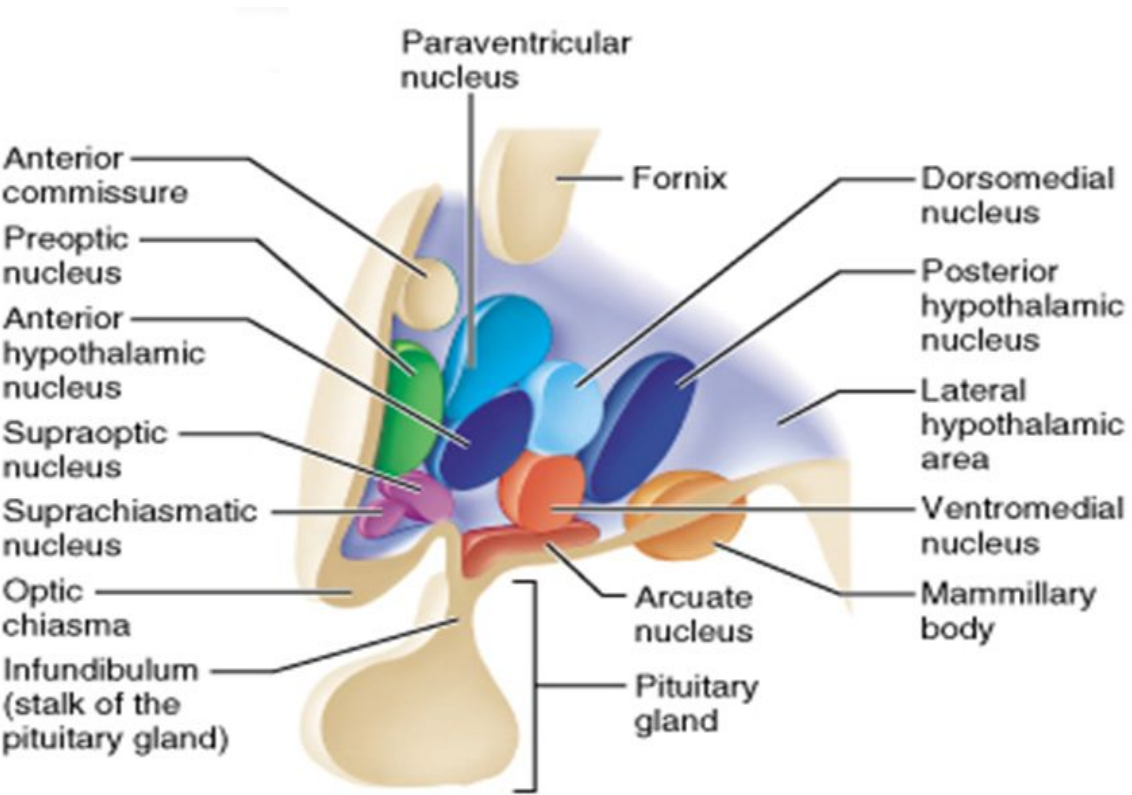


Fig. 13.18. Diagram to show the main hypothalamic nuclei as seen from the medial (ventricular) aspect.

Functions of the Main Hypothalamic Nuclei	
Hypothalamic Nucleus	Function
Supraoptic nucleus	Synthesizes vasopressin (ADH)
Paraventricular nucleus	Synthesizes oxytocin
Preoptic & anterior nuclei	Control parasympathetic system
Posterior & lateral nuclei	Control sympathetic system
Anterior hypothalamic nuclei	Temperature regulation in response to heat
Posterior hypothalamic nuclei	Temperature regulation in response to cold
Lateral hypothalamic nuclei	Hunger center
Medial hypothalamic nuclei	Satiety center
Lateral hypothalamic nuclei	Thirst center
Suprachiasmatic nucleus	Controls circadian rhythms

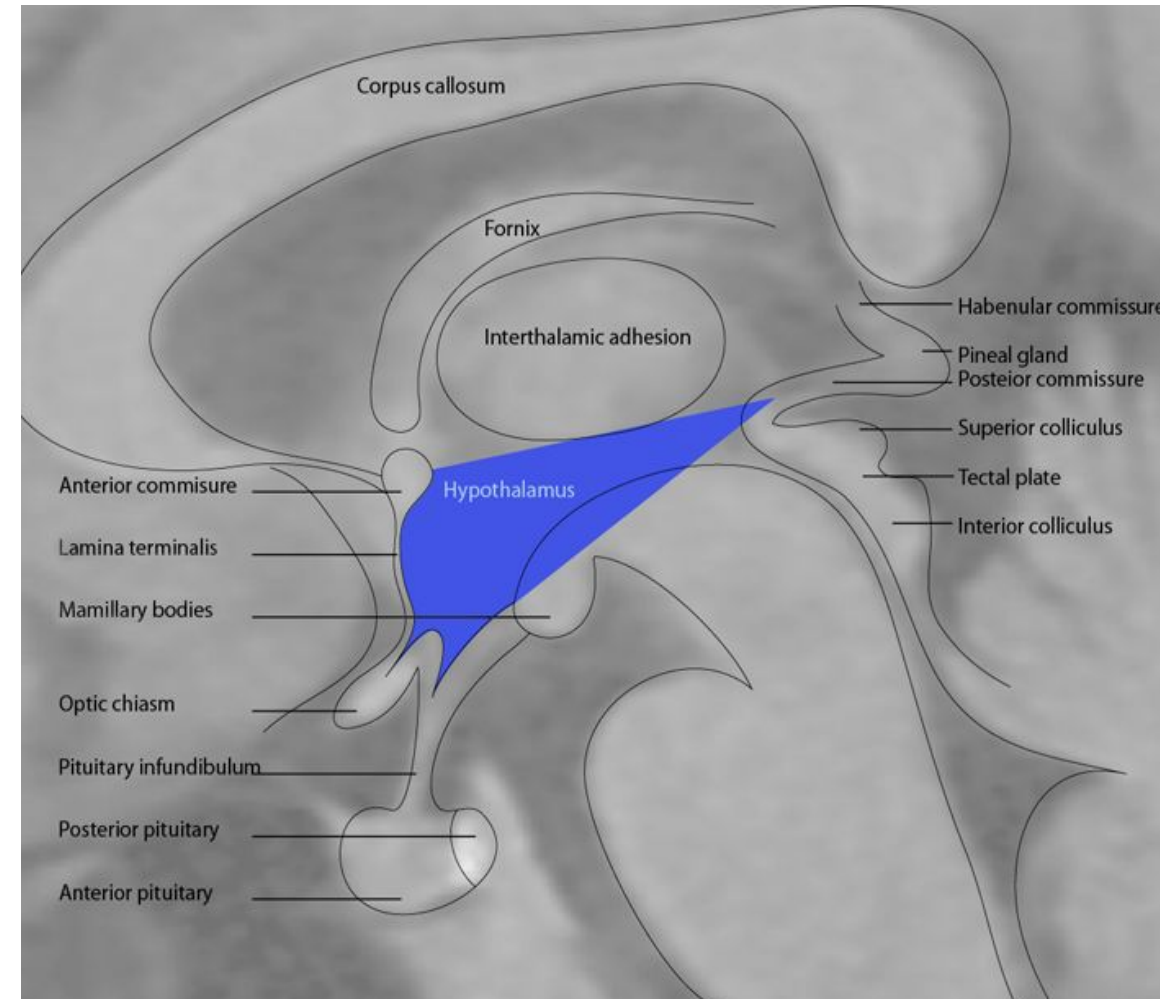


(b)

The Main Afferent and Efferent Nervous Connections of Hypothalamus		
Pathway	Origin	Destination
Afferent		
Medial & spinal lemnisci, tractus solitarius, reticular formation	Viscera & somatic structures	Hypothalamic nuclei
Visual fibers	Retina	Suprachiasmatic nucleus
Medial forebrain bundle	Olfactory mucosa	Hypothalamic nuclei
Auditory fibers	Internal ear	Hypothalamic nuclei
Cortico-hypothalamic fibers	Frontal lobe of cerebral cortex	Hypothalamic nuclei
Hippocampohypothalamic fibers	Hippocampus	Nuclei of mamillary body
Amygdalohypothalamic fibers	Amygdaloid complex	Hypothalamic nuclei
Thalamohypothalamic fibers	Dorsomedial & midline nuclei of thalamus	Hypothalamic nuclei
Tegmental fibers	Tegmentum of midbrain	Hypothalamic nuclei
Efferent		
Descending fibers from reticular formation to brainstem & spinal cord	Preoptic, anterior, posterior & lateral nuclei of hypothalamus	Craniosacral (parasympathetic) & thoracolumbar (sympathetic) outflows
Mammillothalamic tract	Nuclei of mamillary body	Anterior nucleus to thalamus
Mamillotegmental tract	Nuclei of mamillary body	Reticular formation in tegmentum of midbrain

Functions of Hypothalamus

- Controls and integrates activities of the ANS which regulates smooth, cardiac muscle and glands
- Synthesizes regulatory hormones that control the anterior pituitary
- Contains cell bodies of axons that end in posterior pituitary where they secrete hormones
- Regulates rage, aggression, pain, pleasure & arousal
- Feeding, thirst & satiety centers
- Controls body temperature
- Regulates daily patterns of sleep



References:

CLINICAL NEUROANATOMY

Richard S. Snell

8th edition

Chapter 13

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